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# Thermodynamic optimization of the Ho–Sn system

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**Abstract** The thermodynamic optimization of the Ho–Sn binary system was carried out with the help of CALculation of PHAse Diagram (CALPHAD) method. $Ho_5Sn_3$, $Ho_5Sn_4$, $Ho_{11}Sn_{10}$, $Ho_4Sn_5$, $HoSn_2$, $Ho_2Sn_5$, and $HoSn_3$ have been treated as stoichiometric compounds while a solution model has been used for the description of the liquid and HCP_A3 phases. A consistent set of the thermodynamic parameters leading to a reasonable agreement between the calculated results and literature data was obtained.

**Keywords** Ho–Sn system · CALPHAD method · Thermodynamic assessment · Phase diagram

## Introduction

The alloy systems or intermetallic compounds composed of rare earth and Sn have attracted the attention for the magnetic materials [1]. The present work is a part of a thermodynamic description of R–Sn (R = Yb [2] and Sc [3]) already calculated in our laboratory which is required to give a better understanding of the constitutional properties and potential technological applications of these alloys.

By modeling the Ho–Sn alloy, a self-consistent description of the phase relations and thermodynamic data was obtained by means of the CALPHAD technique [4] in the present work.

## Experimental data

The Ho–Sn binary system was investigated by Bulanova and Martsenjuk [5] using thermal and microscopic analysis and the existence of five intermetallic compounds $Ho_5Sn_3$, $Ho_{11}Sn_{10}$, $HoSn_2$, $Ho_2Sn_5$, and $HoSn_3$ was reported. Later, Palenzona and Manfrinetti [6] studied the phase diagram in the Sn-rich region. For the $HoSn_2$, $Ho_2Sn_5$, and $HoSn_3$ compounds, a peritectic formation were reported, respectively, at 1,403, 793, and 693 K.

Recently, by X-ray diffraction and microscopy technique, Bulanova et al. [7] reinvestigated the system; seven intermetallic compounds were reported ($Ho_5Sn_3$, $Ho_5Sn_4$, $Ho_{11}Sn_{10}$, $Ho_4Sn_5$, $HoSn_2$, $Ho_2Sn_5$, and $HoSn_3$). For the $Ho_5Sn_3$ compound, a congruent melting was estimated at 2,188 K, while for the $Ho_5Sn_4$, $Ho_{11}Sn_{10}$, $Ho_4Sn_5$, $HoSn_2$, $Ho_2Sn_5$, and $HoSn_3$ compounds, a peritectic formation was reported, respectively, at 1,993, 1,865, 1,417, 1,387, 782, and 693 K (see Fig. 1). The crystal structures of various phases are reported in Table 1.

Meschel and Kleppa [8] measured the standard enthalpy of formation of $Ho_5Sn_3$ using the differential calorimetry method. Colinet et al. [9] and Witusiewicz et al. [10] calculated the enthalpies of formation for four intermetallic compounds: $Ho_5Sn_3$, $Ho_5Sn_4$, $HoSn_2$, and $HoSn_3$. The enthalpies of formation of $HoSn_2$ and $HoSn_3$ has been measured, respectively, by Lebedev et al. [11] and Yamshchikov et al. [12] using the electromotive force measurements method.

Bulanova et al. [7] measured the enthalpies of dissolution of Ho in liquid Sn at 1,473 K.

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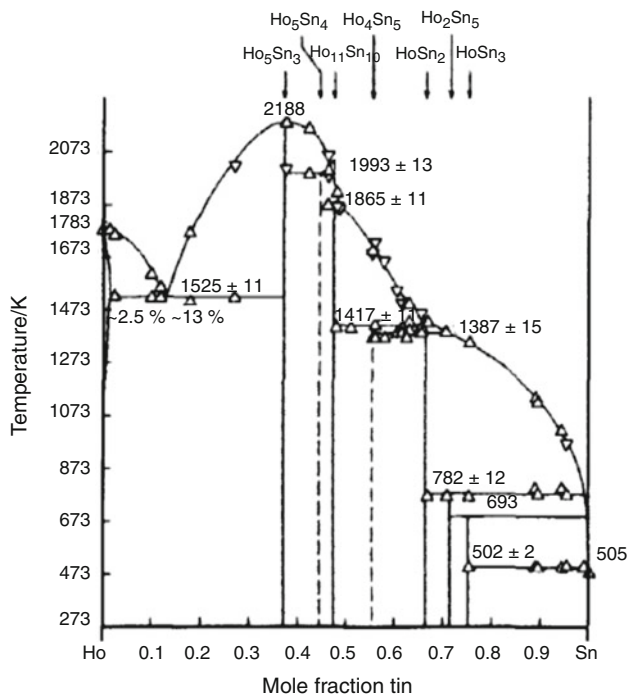


**Fig. 1** The phase diagram of the Ho–Sn system reviewed by Bulanova et al. [7]

## Thermodynamic models

### Pure elements

The Gibbs energy of the pure element $i$ ($i$ = Ho, Sn) in the phase $\varphi$ ($\varphi$ = Liquid, HCP_A3, BCT_A5, and DIAMOND_A4), referred to the enthalpy of its stable state at 298.15 K, is described as a function of temperature by

$$G_i^{\varphi}(T) = {}^0G_i^{\varphi} - H_i^{SER}(298.15\,K) = a + bT + cT\ln T + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9}, \quad (1)$$

where $H_i^{SER}(298.15\,K)$ is the molar enthalpy of the element $i$ at 298.15 K in its standard element reference (SER) state, HCP_A3 for Ho, and BCT_A5 for Sn.

In this paper, The Gibbs energy functions are taken from the SGTE compilation of Dinsdale [13].

### Solution phases

The solution phases, ($\varphi$ = Liquid and HCP_A3) were modeled as substitutional solutions according to the polynomial Redlich–Kister model [14]. The solid BCT_A5 (β-Sn) and Diamond_A4 (α-Sn) with negligible solubilities are considered as pure elements. The Gibbs energy of 1 mol of formula unit of phase $\varphi$ is expressed as the sum of the reference part ${}^{ref}G^{\varphi}$, the ideal part ${}^{id}G^{\varphi}$, and the excess part ${}^{exc}G^{\varphi}$:

$$G_m^{\varphi} = {}^{ref}G^{\varphi} + {}^{id}G^{\varphi} + {}^{exc}G^{\varphi}. \quad (2)$$

As used in the Thermo-Calc software [15]:

$${}^{ref}G^{\varphi}(T) = \left({}^0G_{Ho}^{\varphi}(T) - H_{Ho}^{SER}(298.15\,K)\right)x_{Ho} + \left({}^0G_{Sn}^{\varphi}(T) - H_{Sn}^{SER}(298.15\,K)\right)x_{Sn}, \quad (3)$$

$${}^{id}G^{\varphi} = RT(x_{Ho}\ln x_{Ho} + x_{Sn}\ln x_{Sn}), \quad (4)$$

where $R$ is the gas constant, $T$ is the temperature, in Kelvin, $x_{Ho}$and $x_{Sn}$ are the fraction of elements Ho and Sn, respectively.

The excess ${}^{exc}G^{\varphi}$ energy part in Eq. 3 is given by the Redlich–Kister polynomial [14] formula:

$${}^{exc}G_m^{\varphi}(T) = y_i y_j \sum_{\lambda=0}^{\lambda} {}^{\lambda}L_{i,j}^{\varphi}(y_i - y_j)^{\lambda}. \quad (5)$$

With $i$ and $j$ as indices which correspond to the two species, Holmium and Tin.

The binary interaction parameters of the ${}^{\lambda}L_{i,j}^{\varphi}$ type assessed in this study were temperature dependent as follows:

$${}^{\lambda}L_{i,j}^{\varphi} = a_{\lambda} + b_{\lambda}T \quad (6)$$

$a_{\lambda}$ and $b_{\lambda}$ are the coefficients to be optimized.

**Table 1** Symbols and crystal structures of the stable solid phases in the Ho–Sn system

| Diagram symbol | Composition/at.% Sn | Pearson symbol | Symbol used in Thermo-Calc data file | Prototype |
|---|---|---|---|---|
| Ho | 0 | hP2 | HCP_A3 | Mg |
| $Ho_5Sn_3$ | 37.5 | hP16 | $Ho_5Sn_3$ | $Mn_5Si_3$ |
| $Ho_5Sn_4$ | 44.5 | oP36 | $Ho_5Sn_4$ | $Sm_5Ge_4$ |
| $Ho_{11}Sn_{10}$ | 47.6 | tI84 | $Ho_{11}Sn_{10}$ | $Ho_{11}Ge_{10}$ |
| $Ho_4Sn_5$ | 55.5 | – | $Ho_4Sn_5$ | – |
| $HoSn_2$ | 66.7 | oS12 | $HoSn_2$ | $ZrSi_2$ |
| $Ho_2Sn_5$ | 71.4 | o** | $Ho_2Sn_5$ | $Y_2Sn_5$ |
| $HoSn_3$ | 75 | oS16 | $HoSn_3$ | $GdSn_{2.75}$ |
| β-Sn | 100 | tI4 | BCT_A5 | β-Sn |
| α-Sn | 100 | cF8 | Diamond_A4 | C |

**Table 2** The optimized thermodynamic parameters of the Ho–Sn system

| Phase | Thermodynamic models | Parameters in SI |
|---|---|---|
| Liquid | $(Ho,Sn)_1$ | ${}^0L^{liq} = -202315 - 1.842T$ |
| | | ${}^1L^{liq} = 4573 - 5.985T$ |
| | | ${}^2L^{liq} = 40414 + 13.058T$ |
| HCP_A3 | $(Ho,Sn)_1(Va)_{0.5}$ | ${}^0L^{HCP_A3} = -145966$ |
| BCT_A5 | (Sn) | No excess term |
| DIAMOND_A4 | (Sn) | No excess term |
| $HoSn_3$ | $(Ho)_{0.25}:(Sn)_{0.75}$ | $G^{HoSn3}_{Ho:Sn} - 0.75{}^{298}H^{BCT_A5}_{Sn} - 0.25{}^{298}H^{HCP_A3}_{Ho} = 0.75G^{BCT_A5}_{Sn} + 0.25G^{HCP_A3}_{Ho} - 48951 + 2.751T$ |
| $Ho_2Sn_5$ | $(Ho)_{0.286}:(Sn)_{0.714}$ | $G^{Ho_2Sn_5}_{Sn:Y} - 0.714{}^{298}H^{BCT_A5}_{Sn} - 0.286{}^{298}H^{HCP_A3}_{Ho} = 0.714G^{BCT_A5}_{Sn} + 0.286G^{HCP_A3}_{Ho} - 56034 + 3.739T$ |
| $HoSn_2$ | $(Ho)_{0.333}:(Sn)_{0.667}$ | $G^{HoSn_2}_{Sn:Y} - 0.667{}^{298}H^{BCT_A5}_{Sn} - 0.333{}^{298}H^{HCP_A3}_{Ho} = 0.667G^{BCT_A5}_{Sn} + 0.333G^{HCP_A3}_{Ho} - 65329 + 5.278T$ |
| $Ho_4Sn_5$ | $(Ho)_{0.445}:(Sn)_{0.555}$ | $G^{Ho_4Sn_5}_{Sn:Y} - 0.555{}^{298}H^{BCT_A5}_{Sn} - 0.445{}^{298}H^{HCP_A3}_{Ho} = 0.555G^{BCT_A5}_{Sn} + 0.445G^{HCP_A3}_{Ho} - 71781 + 3.076T$ |
| $Ho_{11}Sn_{10}$ | $(Ho)_{0.524}:(Sn)_{0.476}$ | $G^{Ho_{11}Sn_{10}}_{Sn:Y} - 0.476{}^{298}H^{BCT_A5}_{Sn} - 0.524{}^{298}H^{HCP_A3}_{Ho} = 0.476G^{BCT_A5}_{Sn} + 0.524G^{HCP_A3}_{Ho} - 75965 + 1.727T$ |
| $Ho_5Sn_4$ | $(Ho)_{0.555}:(Sn)_{0.445}$ | $G^{Ho_5Sn_4}_{Sn:Y} - 0.445{}^{298}H^{BCT_A5}_{Sn} - 0.555{}^{298}H^{HCP_A3}_{Ho} = 0.445G^{BCT_A5}_{Sn} + 0.556G^{HCP_A3}_{Ho} - 74370 + 0.762T$ |
| $Ho_5Sn_3$ | $(Ho)_{0.625}:(Sn)_{0.375}$ | $G^{Ho_5Sn_3}_{Sn:Y} - 0.375{}^{298}H^{BCT_A5}_{Sn} - 0.625{}^{298}H^{HCP_A3}_{Ho} = 0.375G^{BCT_A5}_{Sn} + 0.625G^{HCP_A3}_{Ho} - 71087 + .0171T$ |

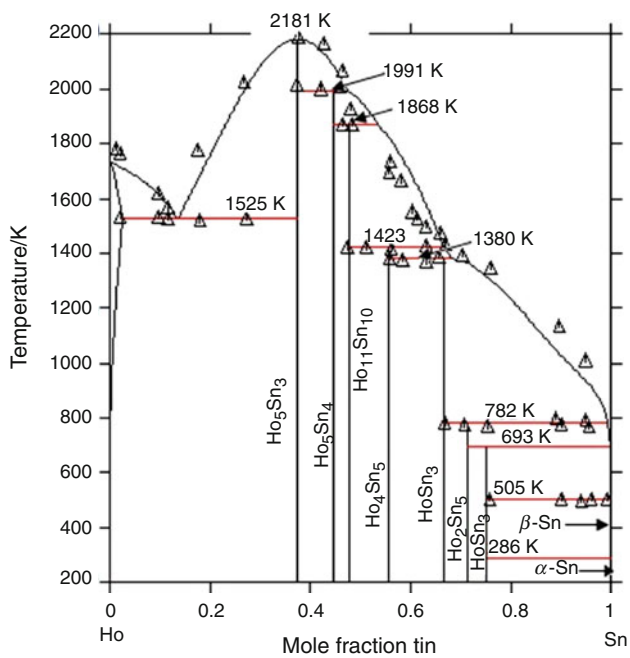


**Fig. 2** The calculated phase diagram of the Ho–Sn system compared with the experimental data [7]

Stoichiometric compounds

The Gibbs energy of the stoichiometric compounds ApBq is expressed as follows:

$$ {}^0G_{A_pB_q} = \frac{p}{p+q}{}^0G_A + \frac{q}{p+q}{}^0G_B + a + bT \quad (7) $$

where ${}^0G_A$ and ${}^0G_B$ are the Gibbs energy of the pure elements Holmium and Tin, respectively, $a$ and $b$ are parameters to be determined.

**Table 3** Invariant reactions in the Ho–Sn system

| Reaction | Bulanova et al. [7] | | This work | |
|---|---|---|---|---|
| | $T$/K | $x^{Liq}_{Sn}$ | $T$/K | $x^{Liq}_{Sn}$ |
| Liq↔ βSn + $HoSn_3$ | 502 | 0.999 | 505 | 1 |
| Liq + $HoSn_3$ ↔ $Ho_2Sn_5$ | 693 | 0.995 | 693 | 0.998 |
| Liq + $Ho_2Sn_5$ ↔ $HoSn_2$ | 782 | 0.992 | 782 | 0.992 |
| Liq + $HoSn_2$ ↔ $Ho_4Sn_5$ | 1,387 | 0.700 | 1,380 | 0.683 |
| Liq + $Ho_4Sn_5$ ↔ $Sn_{10}Ho_{11}$ | 1,417 | 0.660 | 1,423 | 0.666 |
| Liq + $Sn_{10}Ho_{11}$ ↔ $Ho_5Sn_4$ | 1,865 | 0.490 | 1,868 | 0.535 |
| Liq + $Ho_5Sn_4$ ↔ $Ho_5Sn_3$ | 1,993 | 0.460 | 1,991 | 0.474 |
| Liq ↔ $Ho_5Sn_3$ | 2,188 | – | 2,181 | – |
| Liq ↔ Ho +$Ho_5Sn_3$ | 1,525 | 0.130 | 1,525 | 0.135 |

## Results and discussions

The optimization of the thermodynamic parameters was carried out by using the PARROT [16] module of the Thermo-Calc software [15], which is based on a least square procedure. The parameters for the liquid phase were first optimized by the experimental data of phase diagram and thermodynamics of the liquid. The congruent intermetallic compound is going to be investigated next. The other compounds were consequently optimized by using phase diagram data reported by Bulanova et al. [7] and thermodynamic information of the compounds. All the parameters were evaluated and listed in Table 2.

Figure 2 shows the calculated equilibrium phase diagram with all experimental data used in the optimization. A satisfactory agreement is noted in most of the experimental data. The calculated invariant reactions of the Ho–Sn

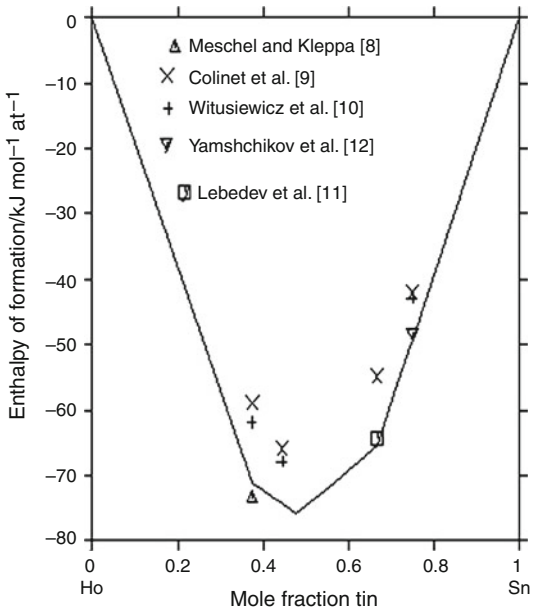


**Fig. 3** Calculated and measured enthalpies of formation of the intermetallic compounds

**Table 4** Calculated and measured enthalpies of formation of the intermetallic compounds

| Phase | $\Delta H_{formation}$/kJ mol$^{-1}$ at$^{-1}$ | Technique used | Reference |
|---|---|---|---|
| $Ho_5Sn_3$ | −73.5 ± 2.3 | Calorimetry | [8] |
| | −59 | Estimation | [9] |
| | −62 ± 5 | Estimation | [10] |
| | −71.1 | Optimization | This work |
| $Ho_5Sn_4$ | −68 ± 6 | Estimation | [10] |
| | −66 | Estimation | [9] |
| | −74.3 | Optimization | This work |
| $HoSn_2$ | −64.3 ± 0.9 | emf | [11] |
| | −55 | Estimation | [9] |
| | −65.3 | Modelization | This work |
| $HoSn_3$ | −48.2 ± 0.65 | Emf | [12] |
| | −42 | Estimation | [9] |
| | −43 ± 6 | Estimation | [10] |
| | −48.9 | Optimization | This work |
| $Ho_2Sn_5$ | −56 | Optimization | This work |
| $Ho_4Sn_5$ | −71.7 | Optimization | This work |
| $Ho_{11}Sn_{10}$ | −75.9 | Optimization | This work |

system are listed in Table 3 and compared with the experimental results from [7].

The assessed enthalpies of formation of the intermetallic compounds compared with experimental measurements are plotted in Fig. 3 and Table 4. The calculated enthalpies agree well with the experimental data [8, 12].

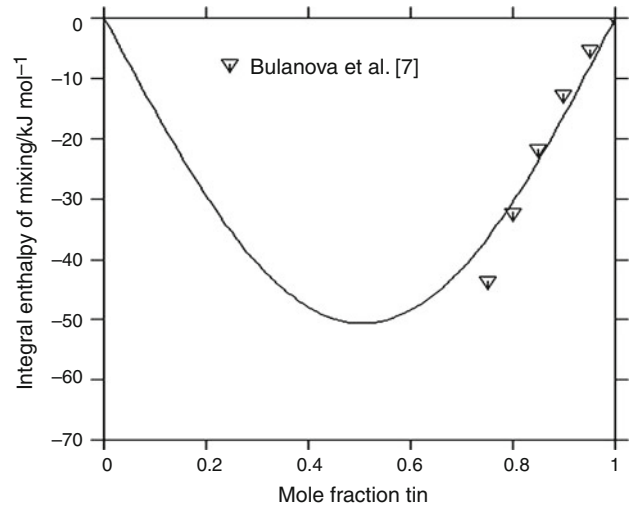


**Fig. 4** Calculated enthalpies of dissolution of Ho in liquid Sn at 1,473 K compared with experimental measurements [7]

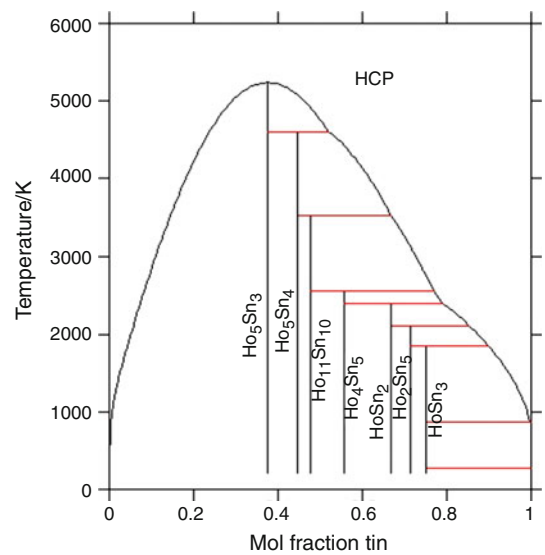


**Fig. 5** Calculated Ho–Sn phase diagram when the liquid phase is suspended

The calculated enthalpies of dissolution of Ho in liquid Sn at 1,473 K are shown in Fig. 4, compared with the experimental value [7]. A satisfactory agreement is observed between our calculation and experimental data.

As mentioned in [17], in order to check that the optimized thermodynamic parameters of the intermetallic compounds are satisfactory, we verified that when the liquid phase is suspended during the calculation of the Ho–Sn phase diagram, the stoichiometric phases disappear at high temperatures. A terminal solid solutions and a two-phase domain existing between them are found to be stable (see Fig. 5).

## Conclusions

The present work reviewed critically the experimental information on phase diagram and thermodynamic properties of the Ho–Sn binary system in published literature. A set of self-consistent thermodynamic parameters formulating the Gibbs energies of various phases in the Ho–Sn binary system was obtained, which can reproduce well most of the experimental data on thermodynamic properties and phase diagram.

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